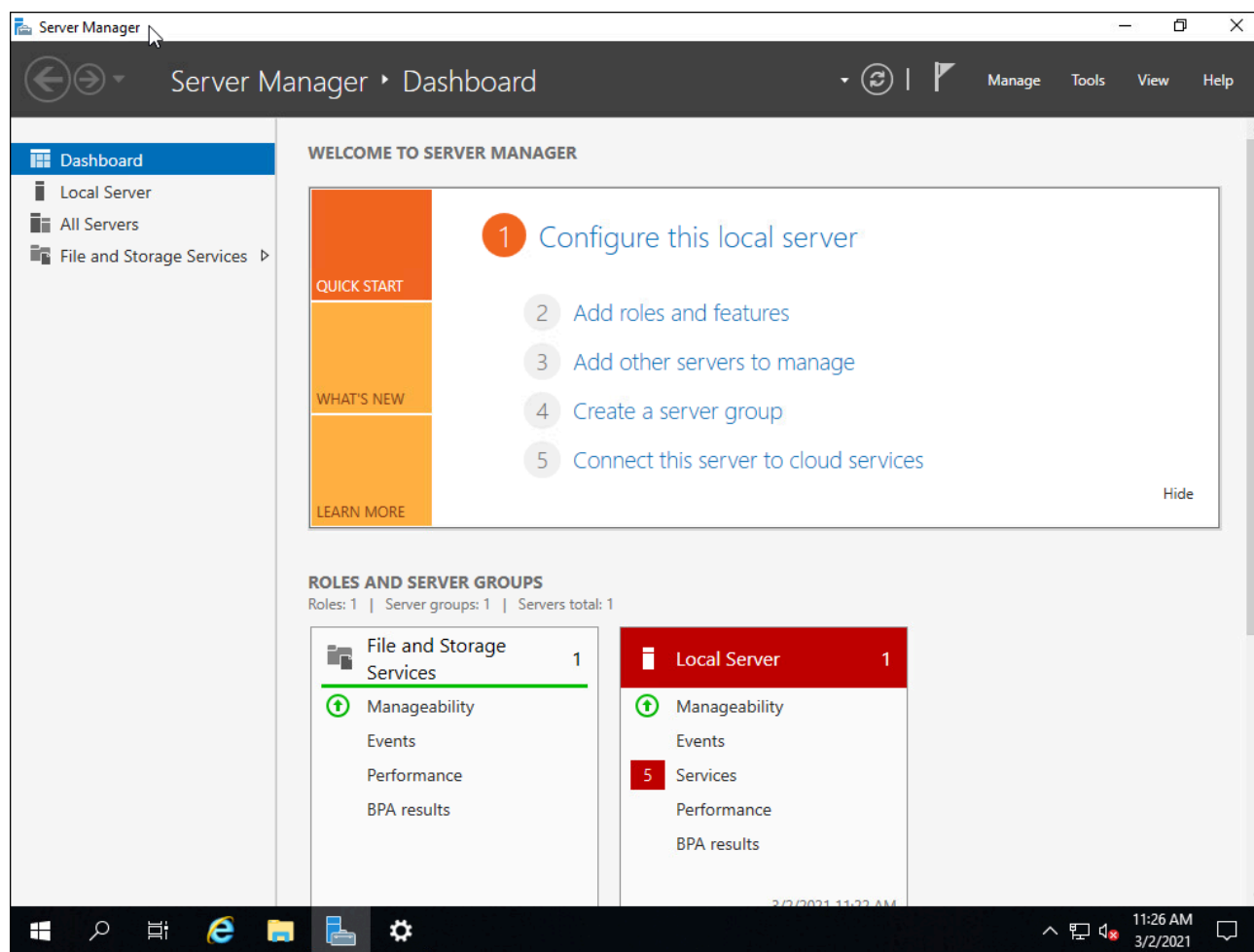


Creating a Red & Blue Team Homelab

hausec.com/2021/03/04/creating-a-red-blue-team-home-lab

By Hausec

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Over the years of penetration testing, red teaming, and teaching, I (and I'm sure a lot of others) are often asked how to get started in infosec. More specifically, how to become a pentester/red teamer or threat hunter/blue teamer. One of the things I always recommend is to build out a lab so you can test TTPs ([techniques](#), [tactics](#), [procedures](#)) and generate IOCs (indicators of compromise) so that you can understand how an attack works and what noise it generates, with the aim of being either to detect that attack or modify it so it's harder to detect. It's not really an opinion, but a matter of fact, that being a better blue teamer will make you a better red teamer and vice-versa. In addition, one of the things that I ask in an interview and have always been asked in an interview, is to describe what your home lab looks like. It's almost an expectation as it is so crucial to be able to experiment with TTPs in a non-production environment. This post is aimed to help you create a home lab that will allow you to both do red team and blue team activity.

Hardware

One of the first questions that's asked about a home lab is the cost. There's a few ways to answer this.

1. Host everything locally on your PC/laptop.
2. Host everything on a dedicated server
3. Host everything in the cloud

The other question is what is the necessary size of the lab? Home-labs do **not** have to replicate the size of an enterprise company. My home lab is setup as shown below, which is what will act as a template for this post.

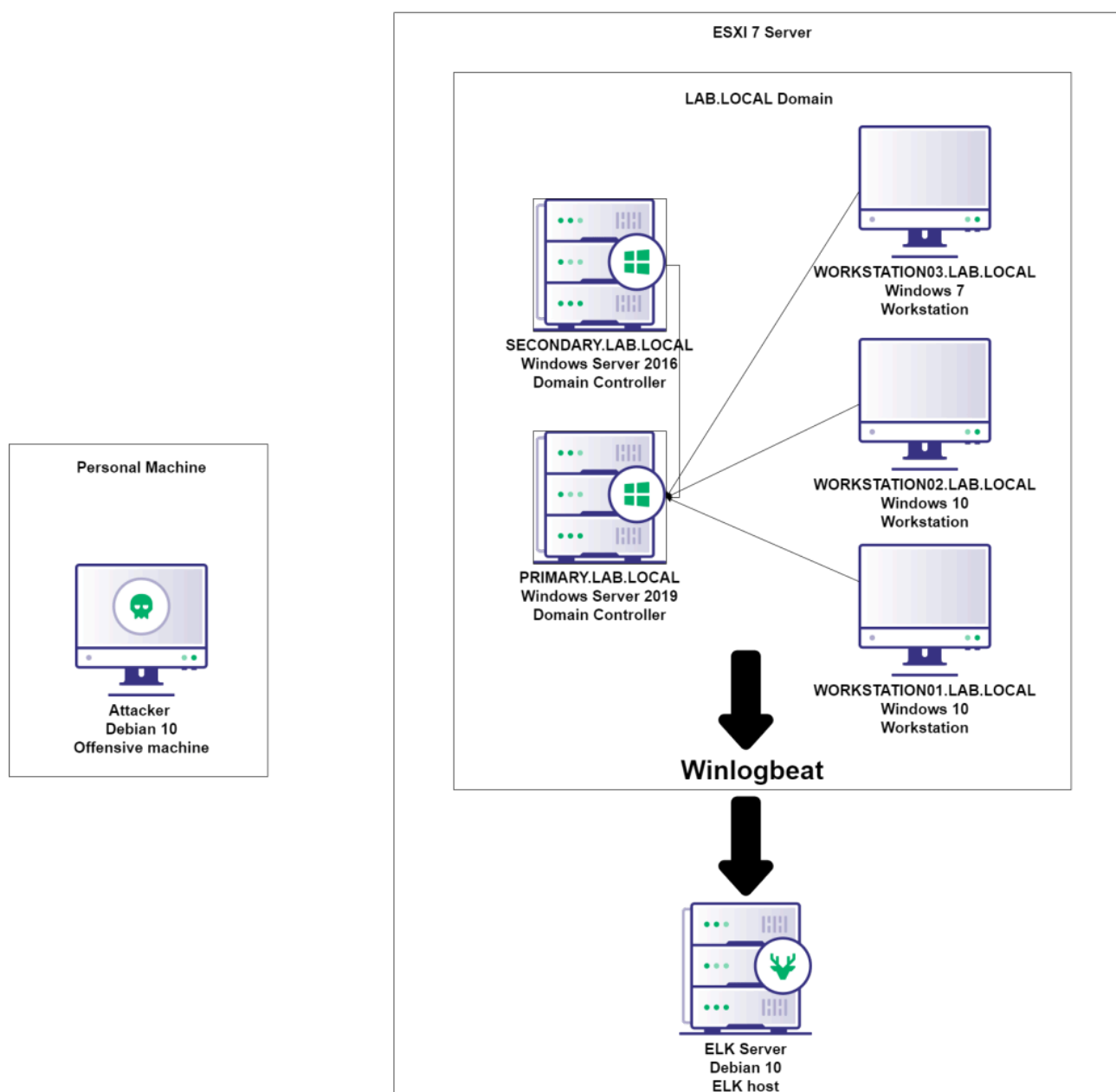


Figure 1: One of many ways to set up a home lab

In my person lab I run two Windows Servers and three Windows workstations. You could absolutely just have one server and one workstation, it's just a matter of what you're trying to accomplish. So, to answer the question of "what will it cost", the answer is "it depends". Personally I use a computer to act as a server which cost me about \$400 to build which runs ESXI 7 to host all the VMs. Cloud could initially be cheaper, but in the long run it will probably cost more. I used to run everything locally on my work PC but I started to run out of disk space with all the VMs. As far as this guide goes, however you choose to host your VMs is up to you.

Hosting OS links:

Server Operating Systems:

[ESXI 7](#)

[Hyper-V](#)

Workstation Applications:

[VMWare Workstation Player](#)

[Virtual Box](#)

Cloud:

[AWS](#)

[Azure](#)

Architecture

How your lab is architected/laid out is a big deal. You want to mimic a real environment as much as possible which is why I suggest building a lab that runs Window's Active Directory (AD). I don't think I've been in an environment where AD was not being used. We will start by using Windows evaluation licenses.

[Windows 10](#)

[Windows Server 2019](#)

And we will use Debian 10 to build an ELK (Elasticsearch, Logstash, Kibana) server.

[Debian 10](#)

Finally, for our attacking machine and for simplicity we will just use Kali

[Kali Linux](#)

ELK Setup

Before setting up Windows, we will set up an ELK server. ELK (Elasticsearch, Logstash, Kibana) is a widely used platform for log processing. As a blue teamer, you want this because digging through logs is a key piece to threat hunting. As a red teamer, you want this to know what IOCs are generated from the TTPs you use.

Keep in mind this lab is meant to be for internal, private use only. The setup of these servers will not be secure and should not be used in a production environment.

Start off by downloading the [Debian 10](#) ISO and then create a VM to boot off the ISO. I won't go into the specifics on creating a VM as it's platform specifics (e.g. VirtualBox, VMWare, etc.), but there's a good article [here](#) for VMWare.

Once you install Debian and log in, you'll want to first add your current user to the sudoers group. First, escalate to root:

```
sudo su
```

Then add your user to the sudoers group.

```
sudo usermod -aG sudo [username]
```

Then switch back to your user

```
su [username]
```

Next, add the GPG key for the Elastic repository

```
wget -q0 - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo apt-key add -
```

And add the repository definition

```
echo "deb https://artifacts.elastic.co/packages/7.x/apt stable main" | sudo tee -a /etc/apt/sources.list.d/elastic-7.x.list
```

Now update apt

```
sudo apt-get update
```

And install logstash

```
sudo apt-get install logstash
```

Then Java

```
sudo apt-get install openjdk-11-jre
```

Then install Elasticsearch

```
sudo apt-get install elasticsearch
```

and finally Kibana

```
sudo apt-get install kibana
```

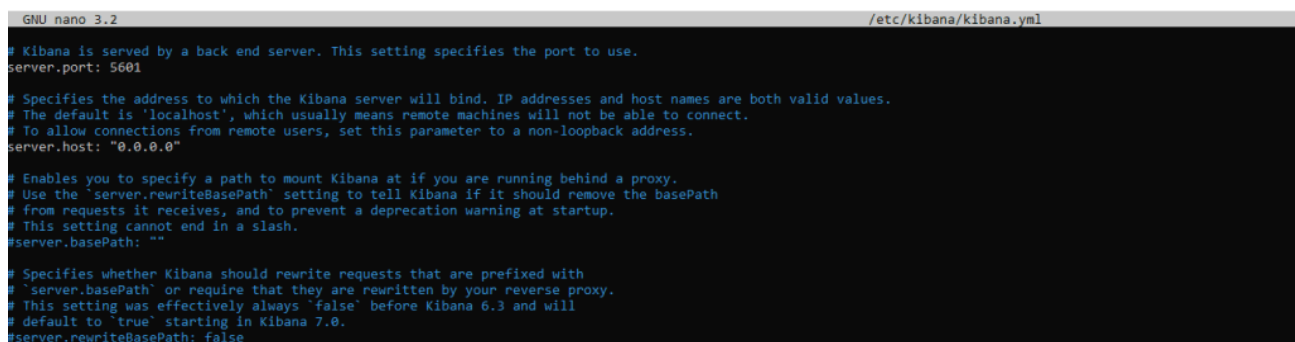
Next is to enable services

```
sudo /bin/systemctl enable elasticsearch.service && sudo /bin/systemctl enable kibana.service && sudo /bin/systemctl enable logstash.service
```

Before we start the services, there's a few config changes we need to make.

```
sudo nano /etc/kibana/kibana.yml
```

Uncomment `server.host` and set the IP to `0.0.0.0` to listen on all interfaces and uncomment `server.port`. You can leave the port to `5601`.



```
GNU nano 3.2 /etc/kibana/kibana.yml
# Kibana is served by a back end server. This setting specifies the port to use.
server.port: 5601

# Specifies the address to which the Kibana server will bind. IP addresses and host names are both valid values.
# The default is 'localhost', which usually means remote machines will not be able to connect.
# To allow connections from remote users, set this parameter to a non-loopback address.
server.host: "0.0.0.0"

# Enables you to specify a path to mount Kibana at if you are running behind a proxy.
# Use the 'server.rewriteBasePath' setting to tell Kibana if it should remove the basePath
# from requests it receives, and to prevent a deprecation warning at startup.
# This setting cannot end in a slash.
server.basePath: ""

# Specifies whether Kibana should rewrite requests that are prefixed with
# 'server.basePath' or require that they are rewritten by your reverse proxy.
# This setting was effectively always 'false' before Kibana 6.3 and will
# default to 'true' starting in Kibana 7.0.
server.rewriteBasePath: false
```

Figure 2: Setting the Kibana Config file

Save the file (ctrl+O, Enter, ctrl+x)

and now edit the elasticsearch config file

```
sudo nano /etc/elasticsearch/elasticsearch.yml
```

Set the `network.host` line to `0.0.0.0` and `http.port` to `9200`

```

GNU nano 3.2 /etc/elasticsearch/elasticsearch.yml
# ----- Elasticsearch Configuration -----
#
# NOTE: Elasticsearch comes with reasonable defaults for most settings.
#       Before you set out to tweak and tune the configuration, make sure you
#       understand what are you trying to accomplish and the consequences.
#
# The primary way of configuring a node is via this file. This template lists
# the most important settings you may want to configure for a production cluster.
#
# Please consult the documentation for further information on configuration options:
# https://www.elastic.co/guide/en/elasticsearch/reference/index.html
#
# ----- Cluster -----
#
# Use a descriptive name for your cluster:
#
#cluster.name: my-application
#
# ----- Node -----
#
# Use a descriptive name for the node:
#
#node.name: node-1
#
# Add custom attributes to the node:
#
#node.attr.rack: r1
#
# ----- Paths -----
#
# Path to directory where to store the data (separate multiple locations by comma):
#
#path.data: /var/lib/elasticsearch
#
# Path to log files:
#
#path.logs: /var/log/elasticsearch
#
# ----- Memory -----
#
# Lock the memory on startup:
#
#bootstrap.memory_lock: true
#
# Make sure that the heap size is set to about half the memory available
# on the system and that the owner of the process is allowed to use this
# limit.
#
# Elasticsearch performs poorly when the system is swapping the memory.
#
# ----- Network -----
#
# Set the bind address to a specific IP (IPv4 or IPv6):
#
#network.host: 0.0.0.0
#
# Set a custom port for HTTP:
#
#http.port: 9200
#
# For more information, consult the network module documentation.

```

Figure 3: Setting the Elasticsearch config settings

And add an additional line at the bottom

discovery.type: single-node

```

http.port: 9200
#
# For more information, consult the network module documentation.
#
# ----- Discovery -----
#
# Pass an initial list of hosts to perform discovery when this node is started:
# The default list of hosts is ["127.0.0.1", "[::1]"]
#
#discovery.seed_hosts: ["host1", "host2"]
#
# Bootstrap the cluster using an initial set of master-eligible nodes:
#
#cluster.initial_master_nodes: ["node-1", "node-2"]
#
# For more information, consult the discovery and cluster formation module documentation.
discovery.type: single-node
#
# ----- Gateway -----
#
# Block initial recovery after a full cluster restart until N nodes are started:
#
#gateway.recover_after_nodes: 3
#
# For more information, consult the gateway module documentation.
#
# ----- Various -----
#

```

Save the file (ctrl+O, Enter, ctrl+x)

And start the services

```
sudo service logstash start
```

```
sudo service elasticsearch start
```

```
sudo service kibana start
```

Now if you browse to your Debian machine's IP on port 5601 you should see Kibana.

```
ip addr
```

```
hausec@elk:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:08:e8:23 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.250/24 brd 192.168.1.255 scope global noprefixroute ens32
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe08:e823/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Figure 4: Viewing the host's IP

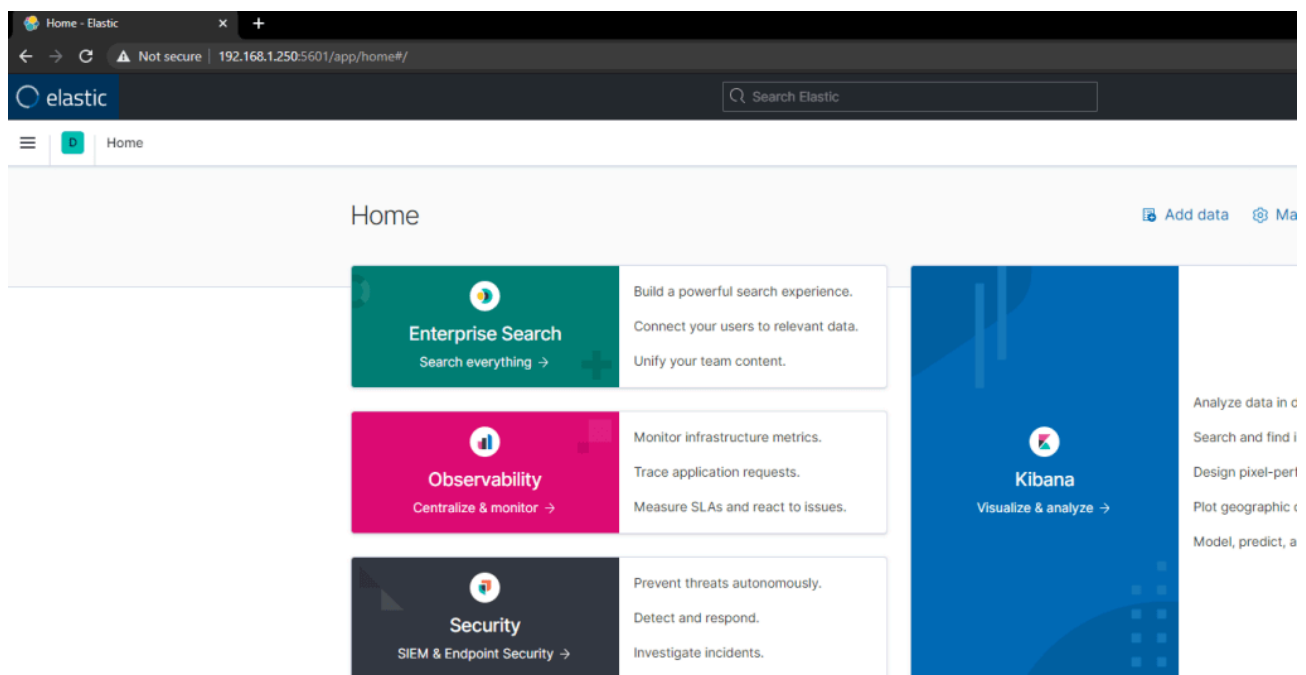


Figure 5: Kibana/Elastic homepage

Windows Setup

Once again, I will not be showing how to deploy a VM as I want this post to be platform agnostic. So for setting up Windows, this will be under the impression you have stood up a Windows 2019 Server and Windows 10 machine.

In this section we will create an Active Directory lab by making a Domain Controller and Workstation.

Windows Server 2019

Once Server 19 is stood up, the first thing you should do is set a static IP. If you don't, the machine's IP can change which will break the environment. For reference, these are my settings.

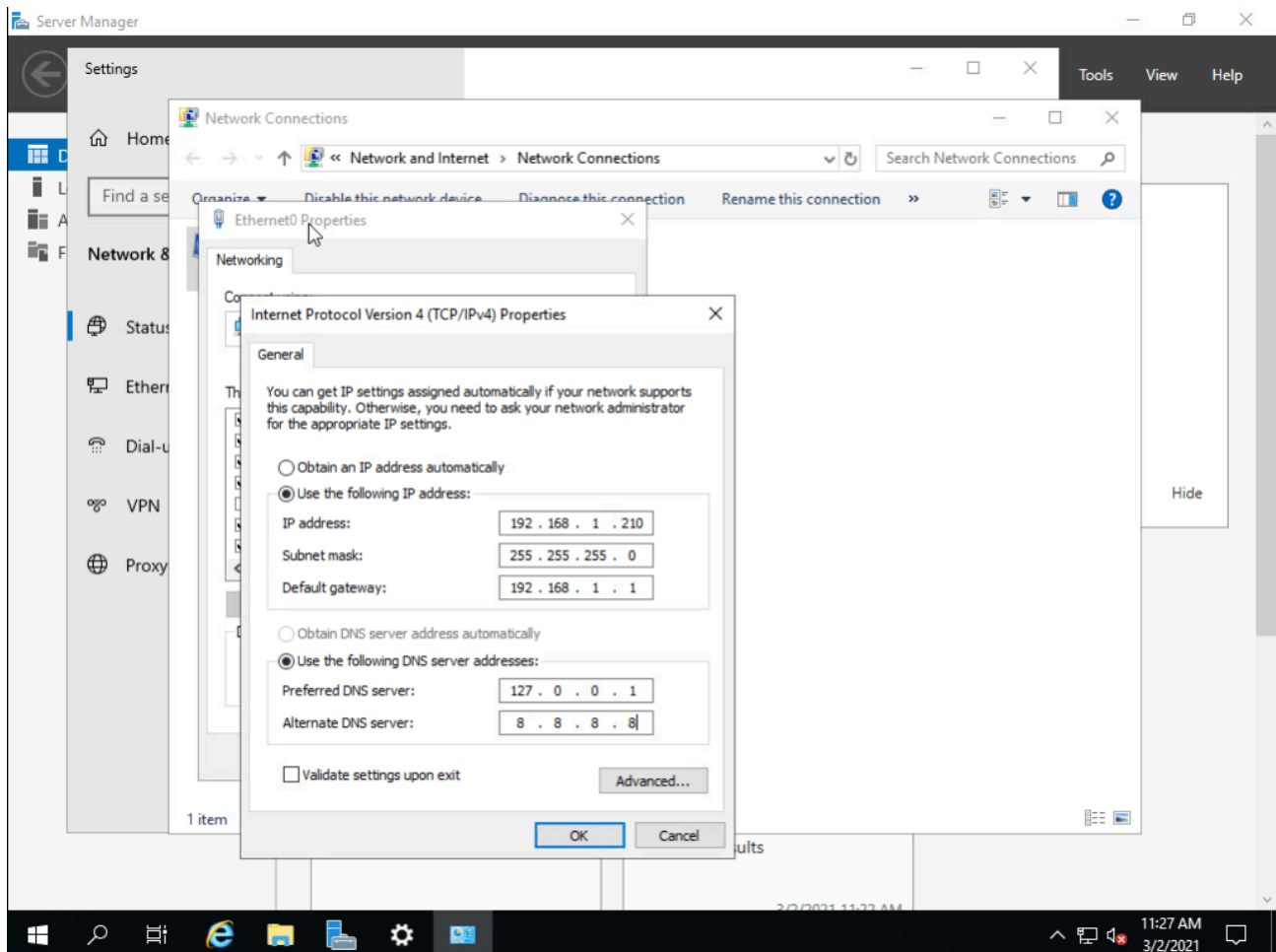


Figure 6: Server's IP settings

The import part here is setting the DNS servers. The preferred DNS will be localhost (127.0.0.1) as we will install the DNS service on the machine in a moment. Then setting Google's DNS server as a secondary so it can reach the internet (optional, completely OK if you do not want your lab to reach the internet).

Next, rename the server to something more conventional. I named mine PRIMARY as it will act as the primary domain controller in the environment.

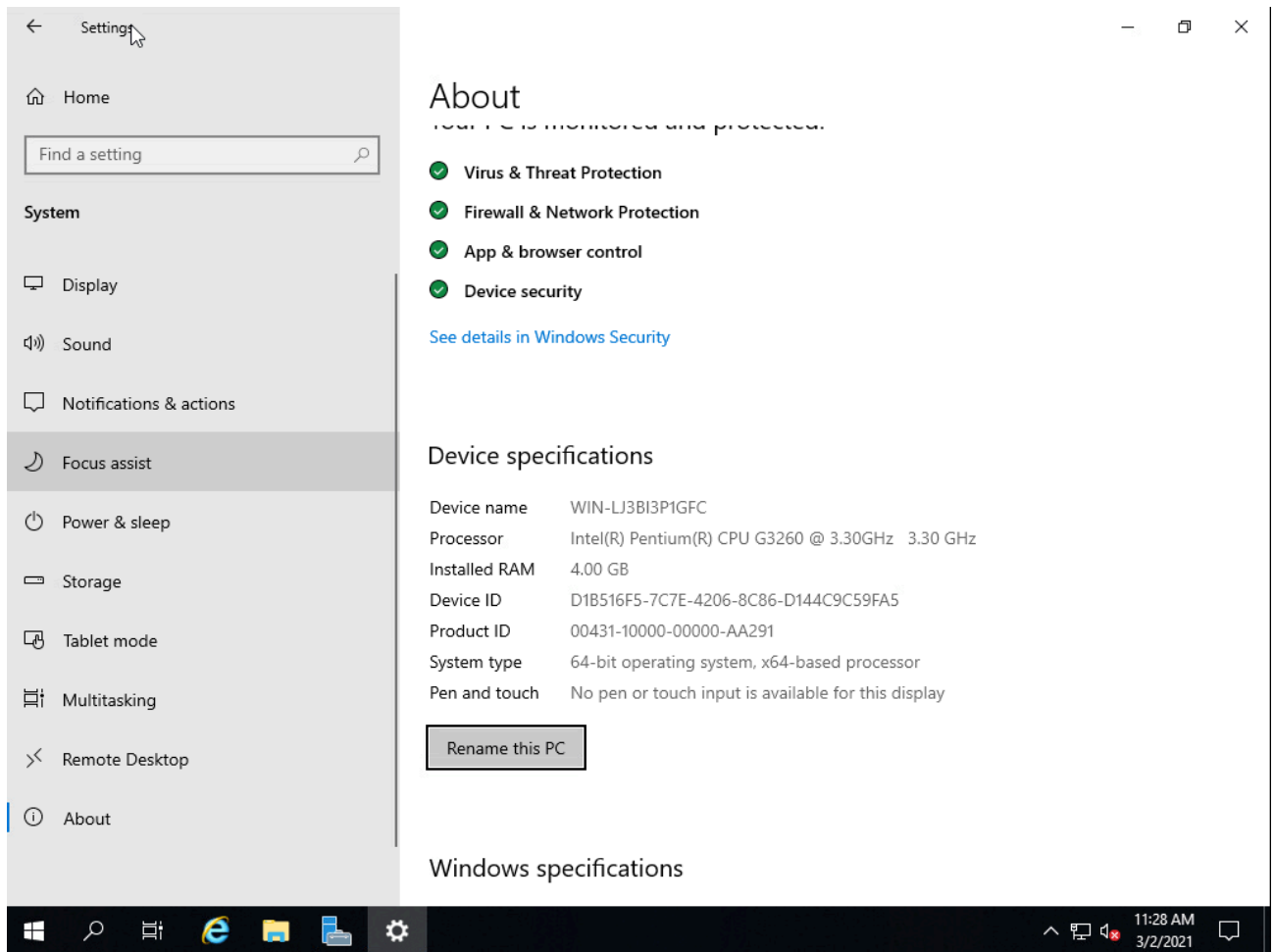


Figure 7: Go to Start>Type in “rename” and this is the screen that will be brought up

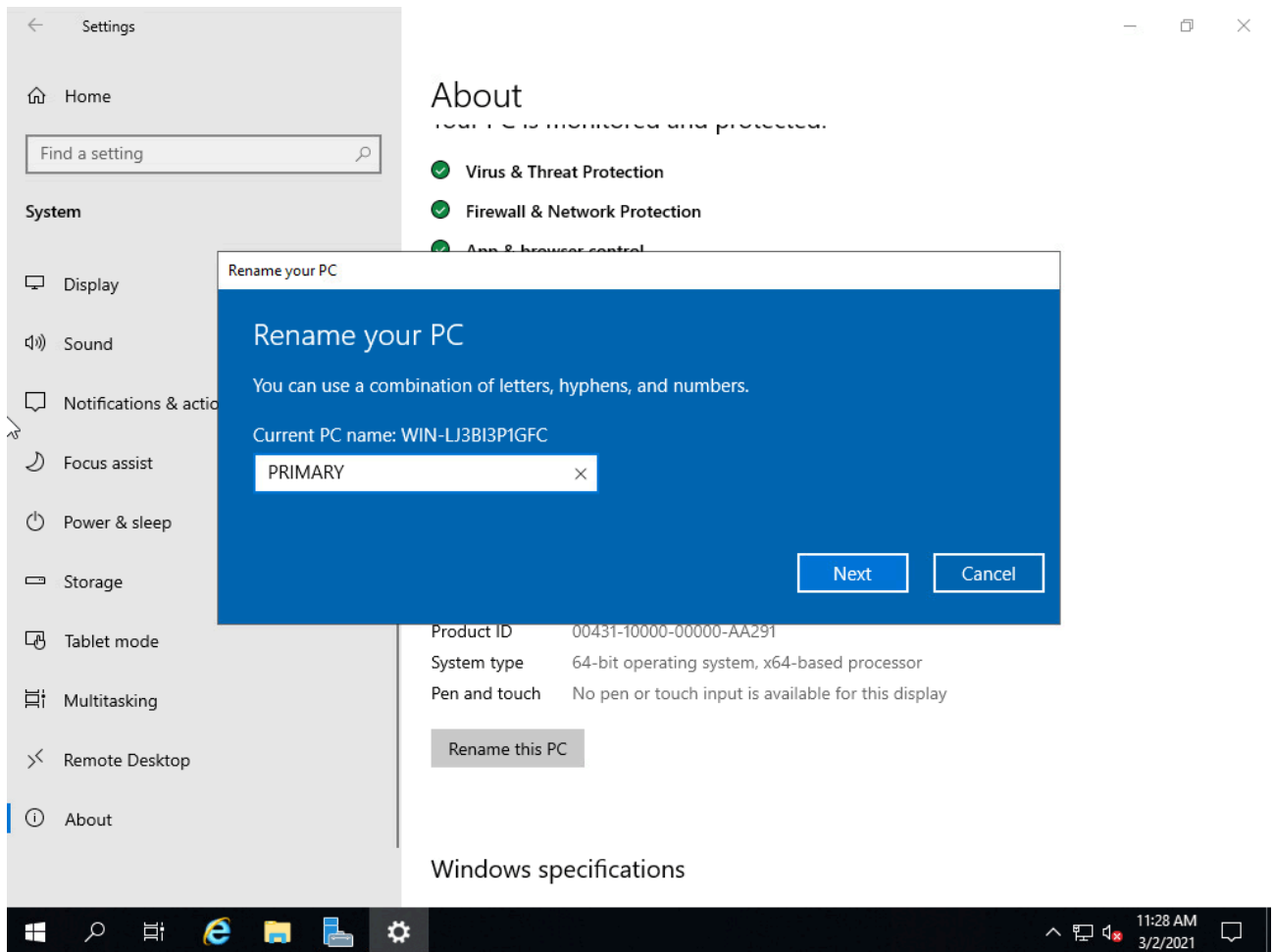


Figure 8: Renaming the server to PRIMARY

Reboot the server for the new settings to take effect. Once rebooted, you should have the server manager dashboard. Click on 'Add roles and features'

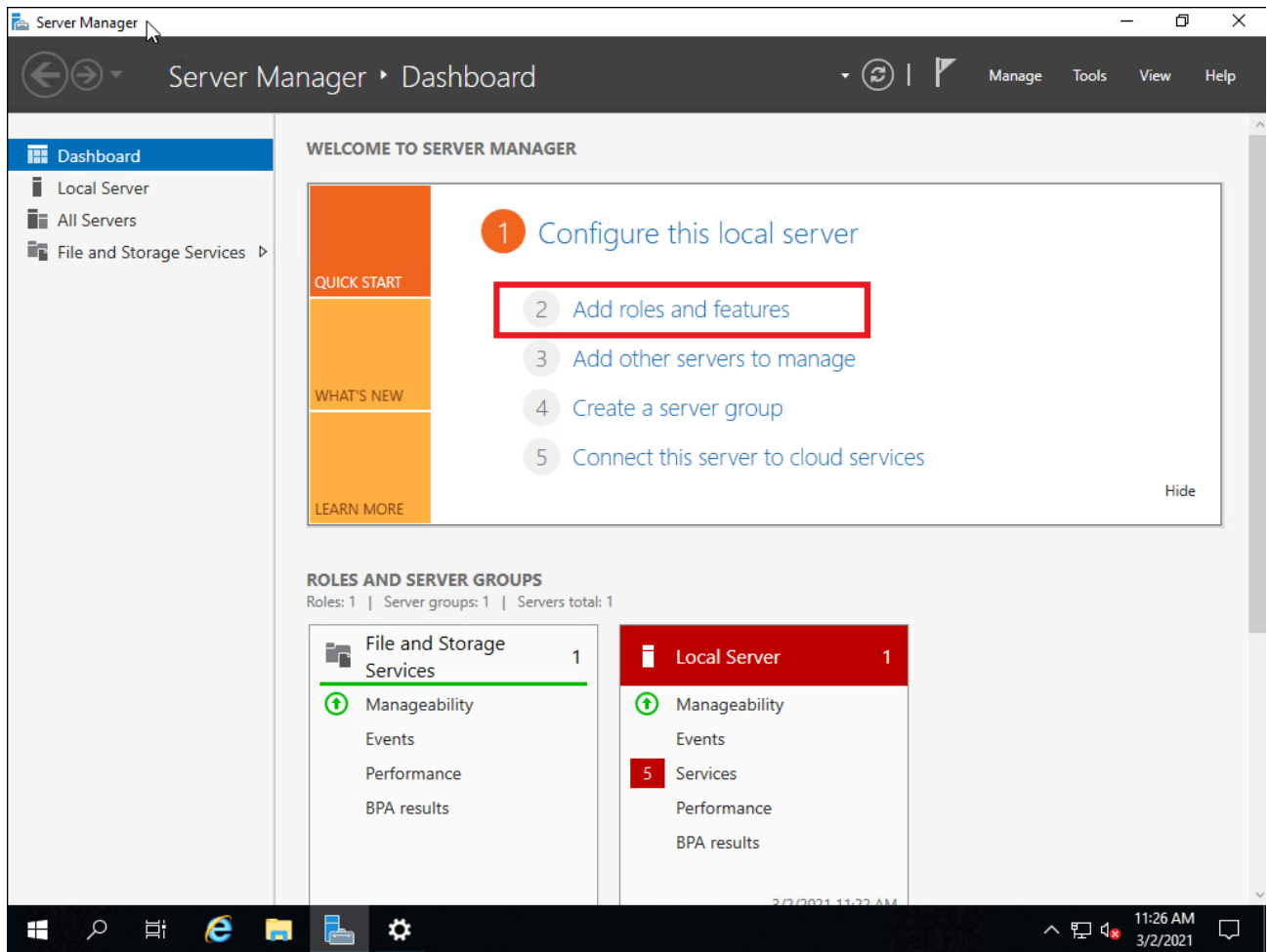


Figure 9: Server Manager Dashboard

Click next until you get to 'Server Roles'. Add DNS and Active Directory Domain Servers

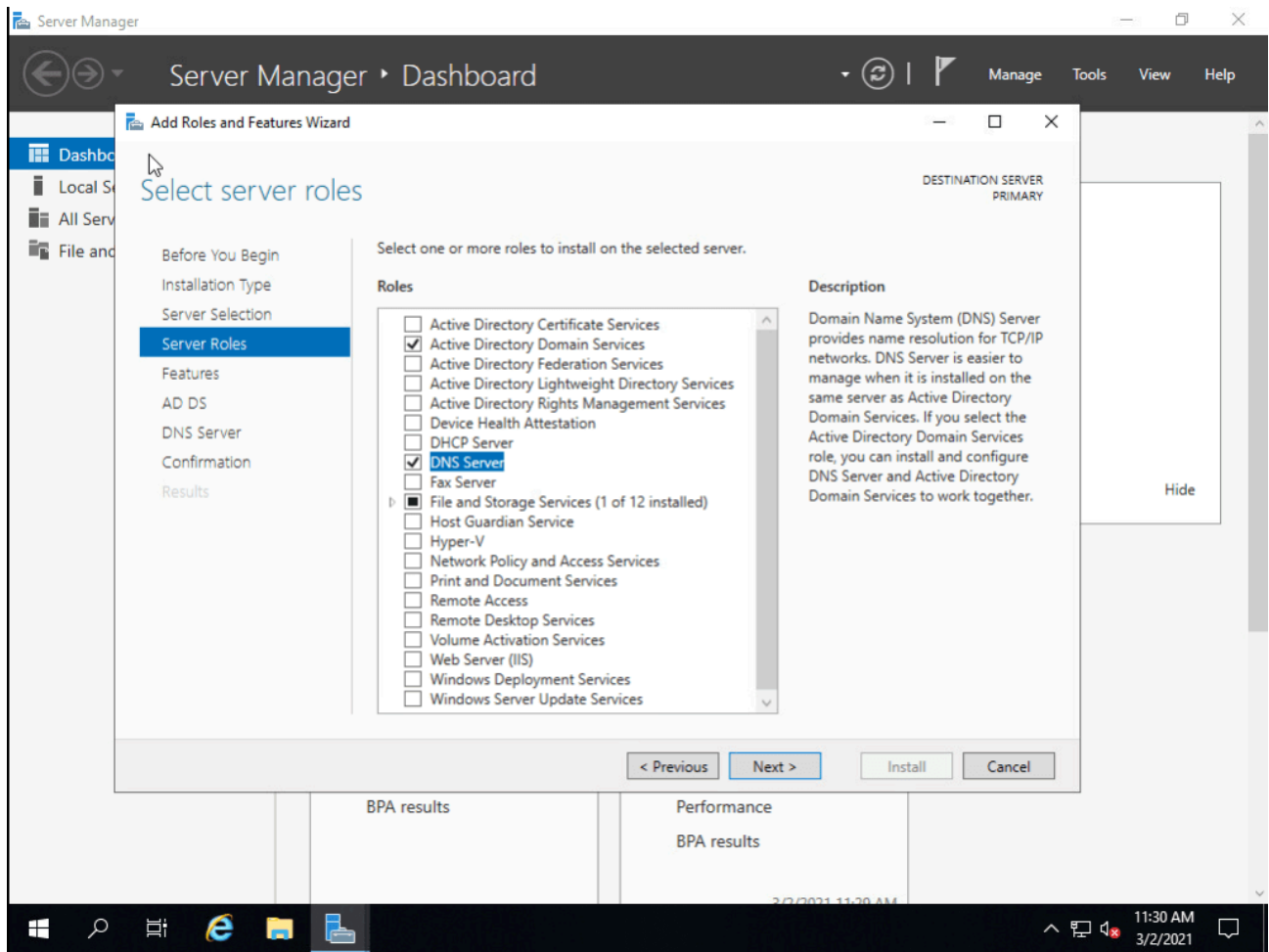


Figure 10: Adding DNS and ADDS services

Click next until it asks for confirmation, then click install.

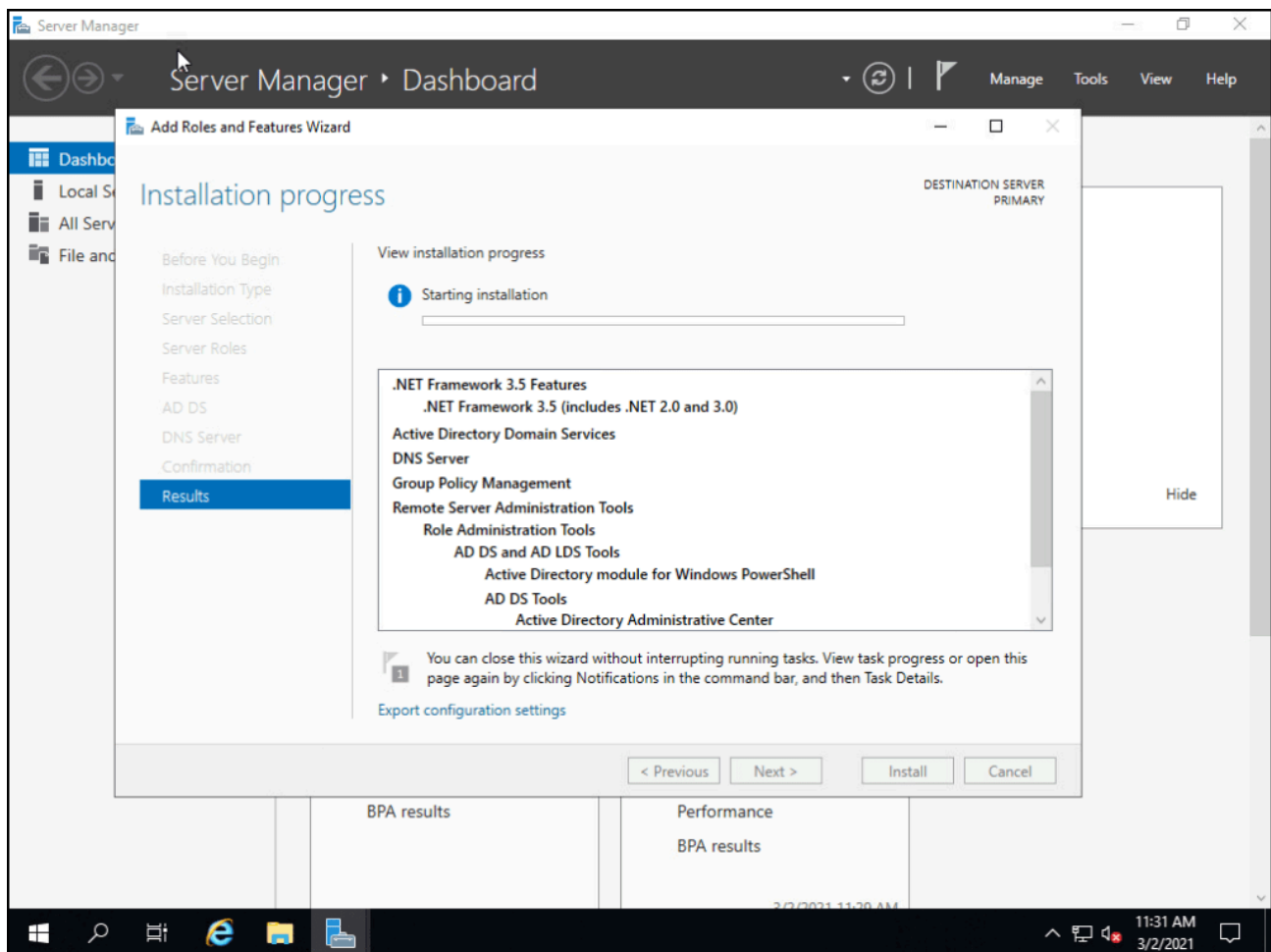


Figure 11: Installing features

After it installs, the server dashboard will have a notification. Click on it and click 'Promote this server to a domain controller'

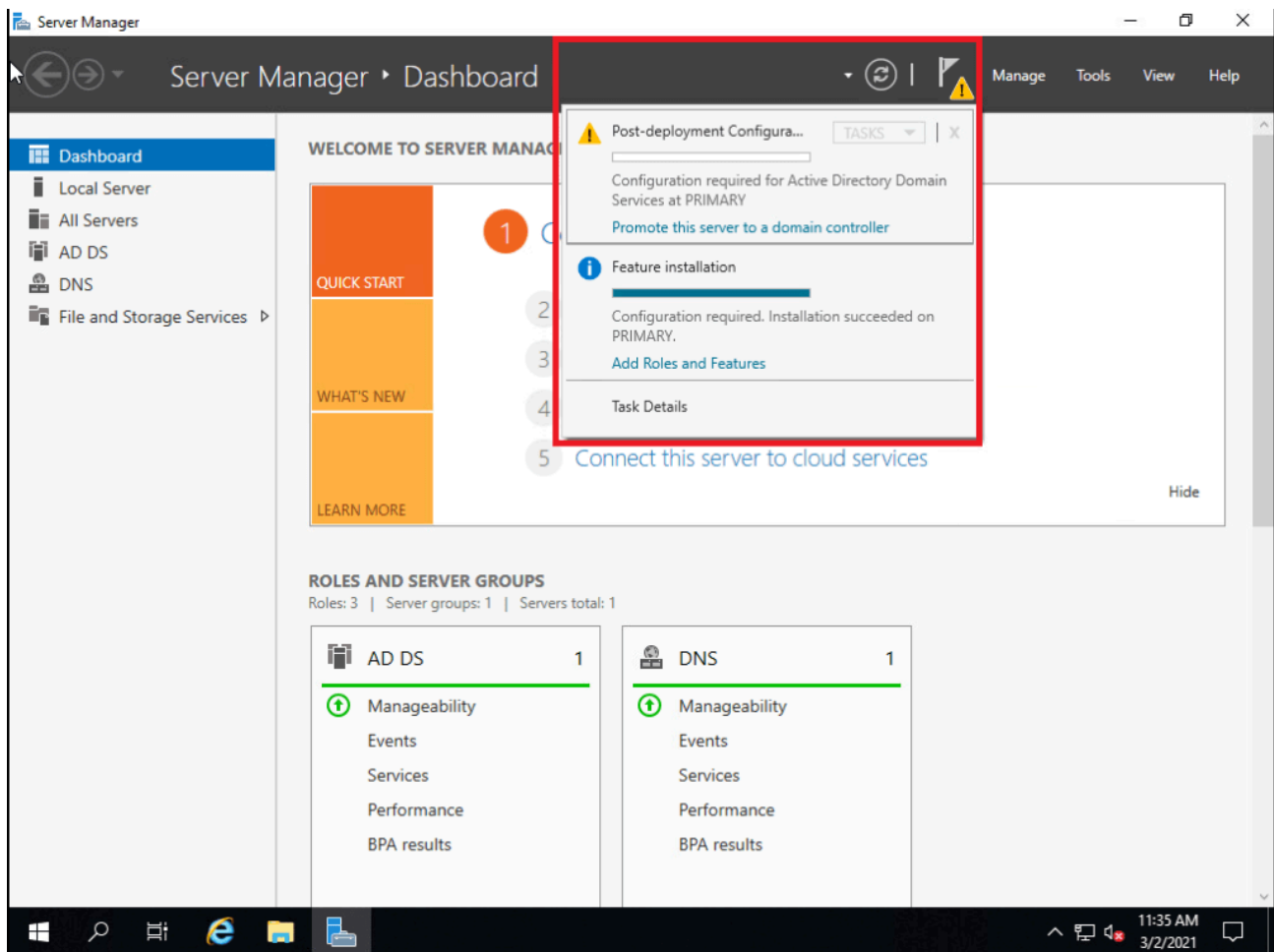


Figure 12: Promote the machine to a DC

Once you click promote, it will bring up another window. Click 'Add a new forest' and give the domain a name. I named mine 'LAB.LOCAL'

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the Windows logo, the text 'Active Directory Domain Services Configuration Wizard', and standard window controls (minimize, maximize, close). The main window has a light gray background. On the left, a vertical sidebar contains a list of steps: 'Deployment Configuration' (highlighted with a blue background), 'Domain Controller Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main area is titled 'Deployment Configuration' in a large blue font. In the top right corner, it says 'TARGET SERVER PRIMARY'. Below the title, there are three radio button options under the heading 'Select the deployment operation': 'Add a domain controller to an existing domain', 'Add a new domain to an existing forest', and 'Add a new forest' (which is selected with a black dot). Below these options, under the heading 'Specify the domain information for this operation', there is a text box labeled 'Root domain name:' containing the text 'LAB.LOCAL'. At the bottom of the main area, there is a blue link that says 'More about deployment configurations'. At the very bottom of the window, there is a gray bar containing four buttons: '< Previous', 'Next >' (highlighted with a blue border), 'Install', and 'Cancel'.

Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
PRIMARY

Deployment Configuration

- Domain Controller Options
- Additional Options
- Paths
- Review Options
- Prerequisites Check
- Installation
- Results

Select the deployment operation

- ☐ Add a domain controller to an existing domain
- ☐ Add a new domain to an existing forest
- ☒ Add a new forest

Specify the domain information for this operation

Root domain name:

[More about deployment configurations](#)

< Previous Next > Install Cancel

Figure 13: Give your domain a name

Next, leave the default functional levels (Unless you're adding a 2012, 2008, or 2003 server, then change it to those). Then set the DSRM password to something you'll remember.

Active Directory Domain Services Configuration Wizard

Domain Controller Options

TARGET SERVER
PRIMARY

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2016

Domain functional level: Windows Server 2016

Specify domain controller capabilities

- ☒ Domain Name System (DNS) server
- ☒ Global Catalog (GC)
- ☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

[More about domain controller options](#)

< Previous Next > Install Cancel

Figure 14: Setting the DSRM password

Click next until you get to the prerequisite check, then click 'install'.

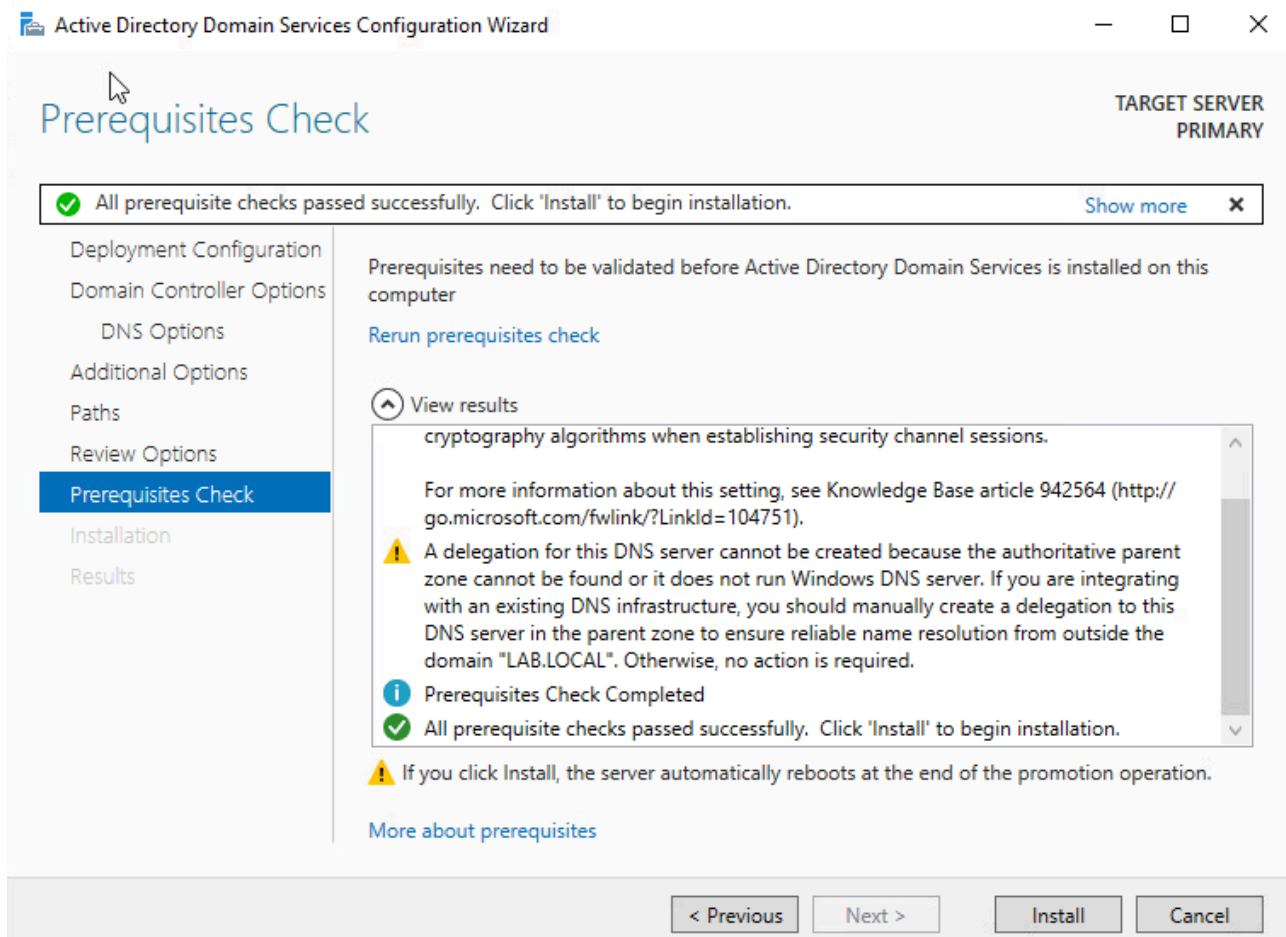


Figure 15: Prereq check will give warnings

Once done, reboot the server.

Once rebooted, in the Server Dashboard, click on Tools>ADUC (Active Directory Users and Computers)

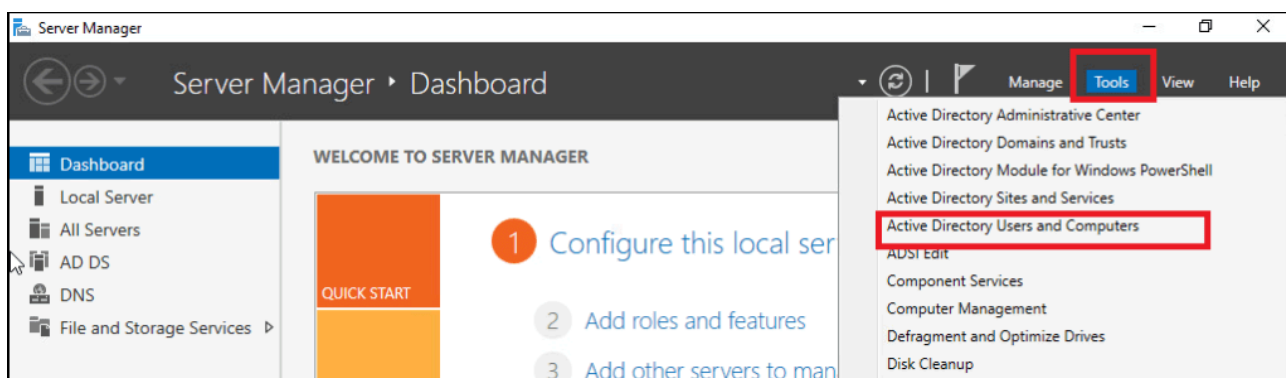


Figure 16: Tools>ADUC

ADUC is used to manage users, groups, and computers (among other things). In this instance we just want to create a new user and assign them to the Domain Administrator role.

In ADUC, click on your domain on the left then select 'Users'. At the top, click the icon shown below to create a new user.

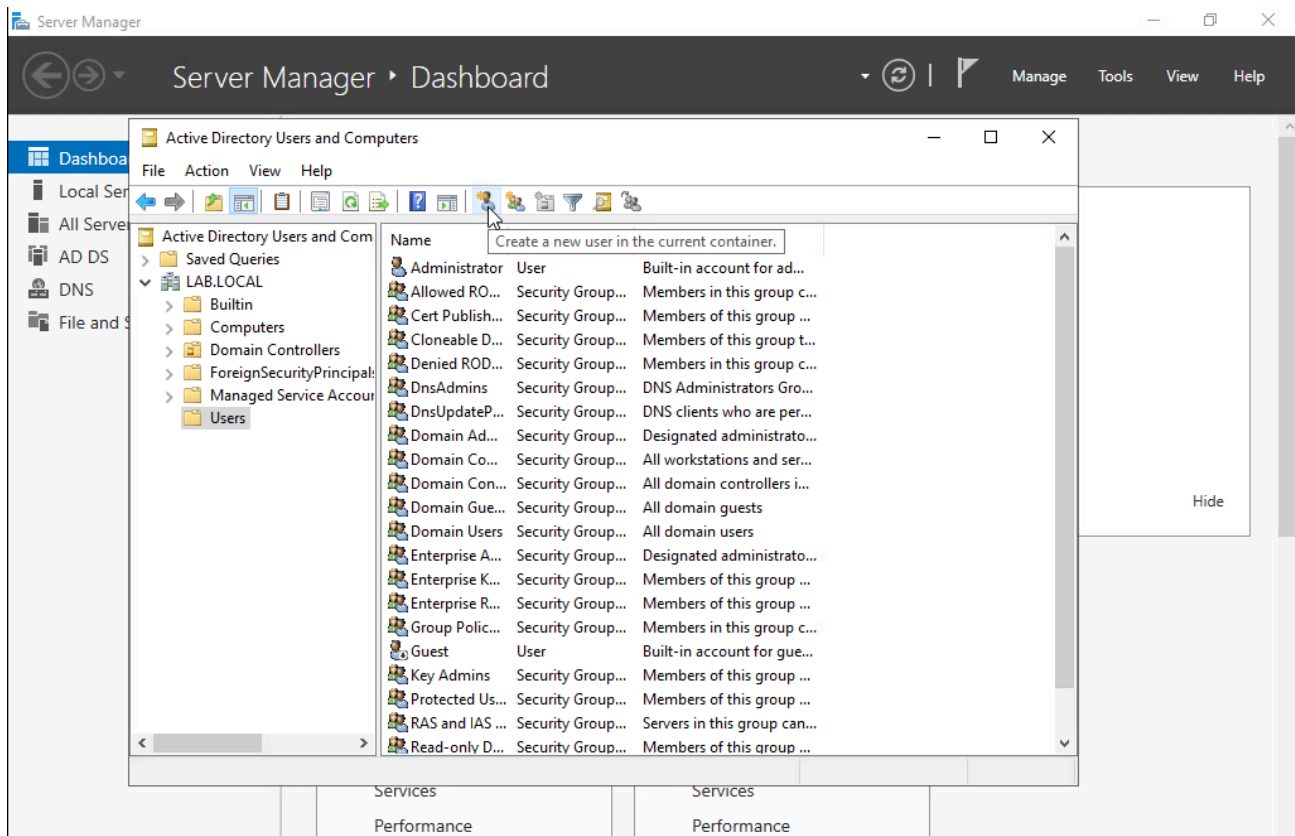


Figure 17: Creating a new user in ADUC

Give them a name where you can identify them as an administrator by their name. Commonly in environments, they have ADM, ADMIN, -A, or some moniker to signify it's a privileged account. Once created, right click on that user in ADUC and click 'Add to a group'

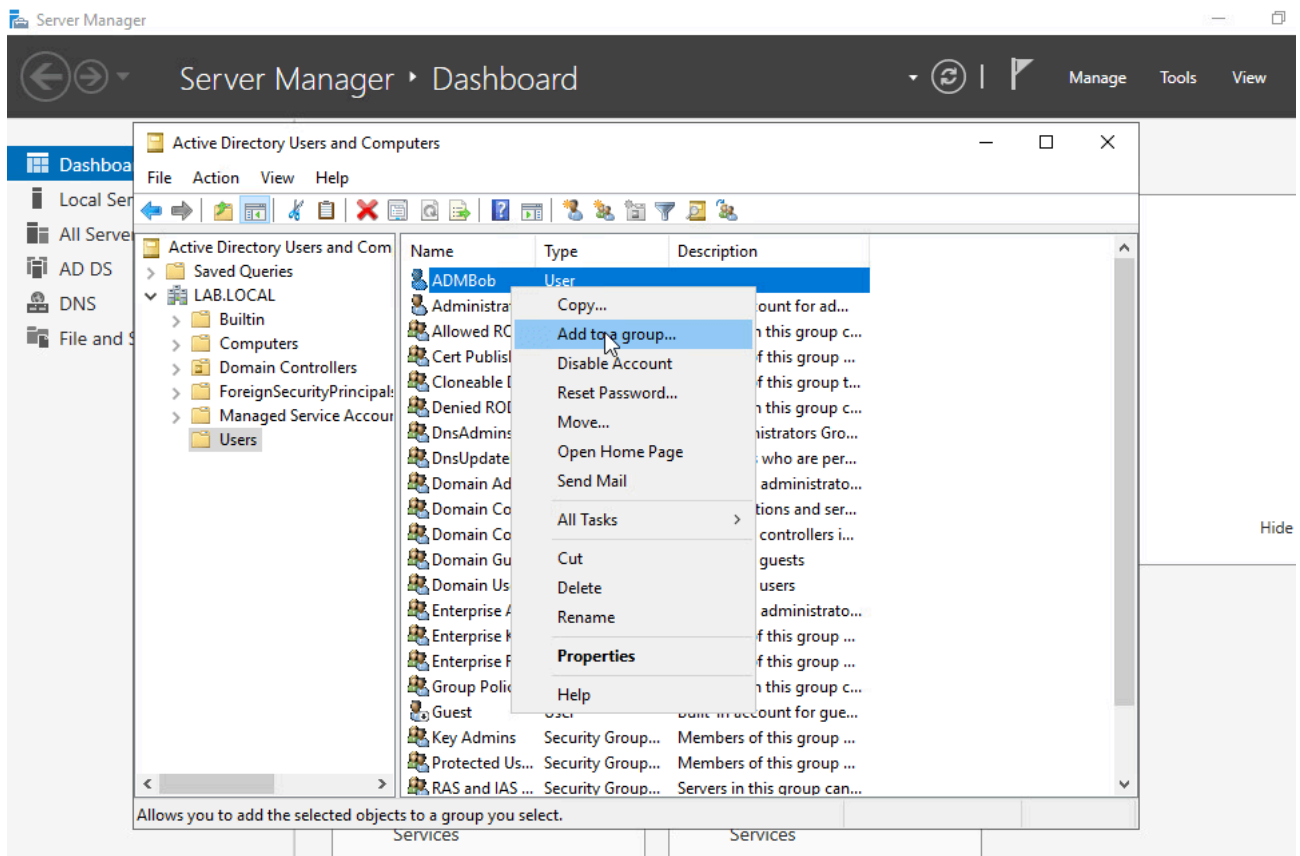


Figure 18: Adding the user to a group

Then type in 'Domain Admins' and select 'OK'.

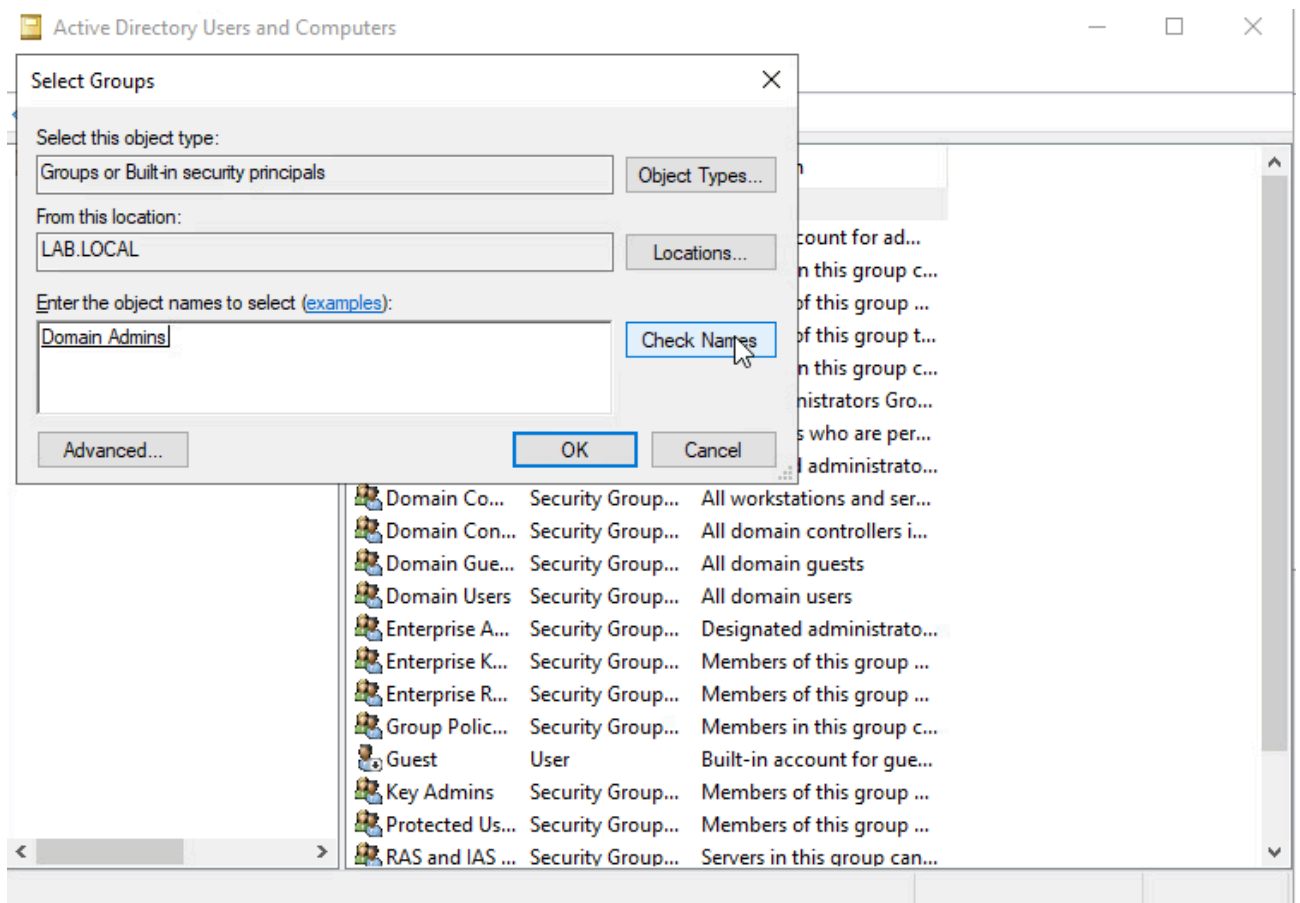


Figure 19: Adding the user to the Domain Admins group

Once the user is added to the Domain Admins group, switch over to the Windows 10 workstation. Once again, I will assume the provisioning of the machine was already done and it is able to communicate with the domain controller. A simple test is to ping the Domain Controller's IP and ensure they can talk to each other on the network.

On the Windows 10 machine, edit the DNS settings to include your Domain Controller's IP address. Below is a shown example.

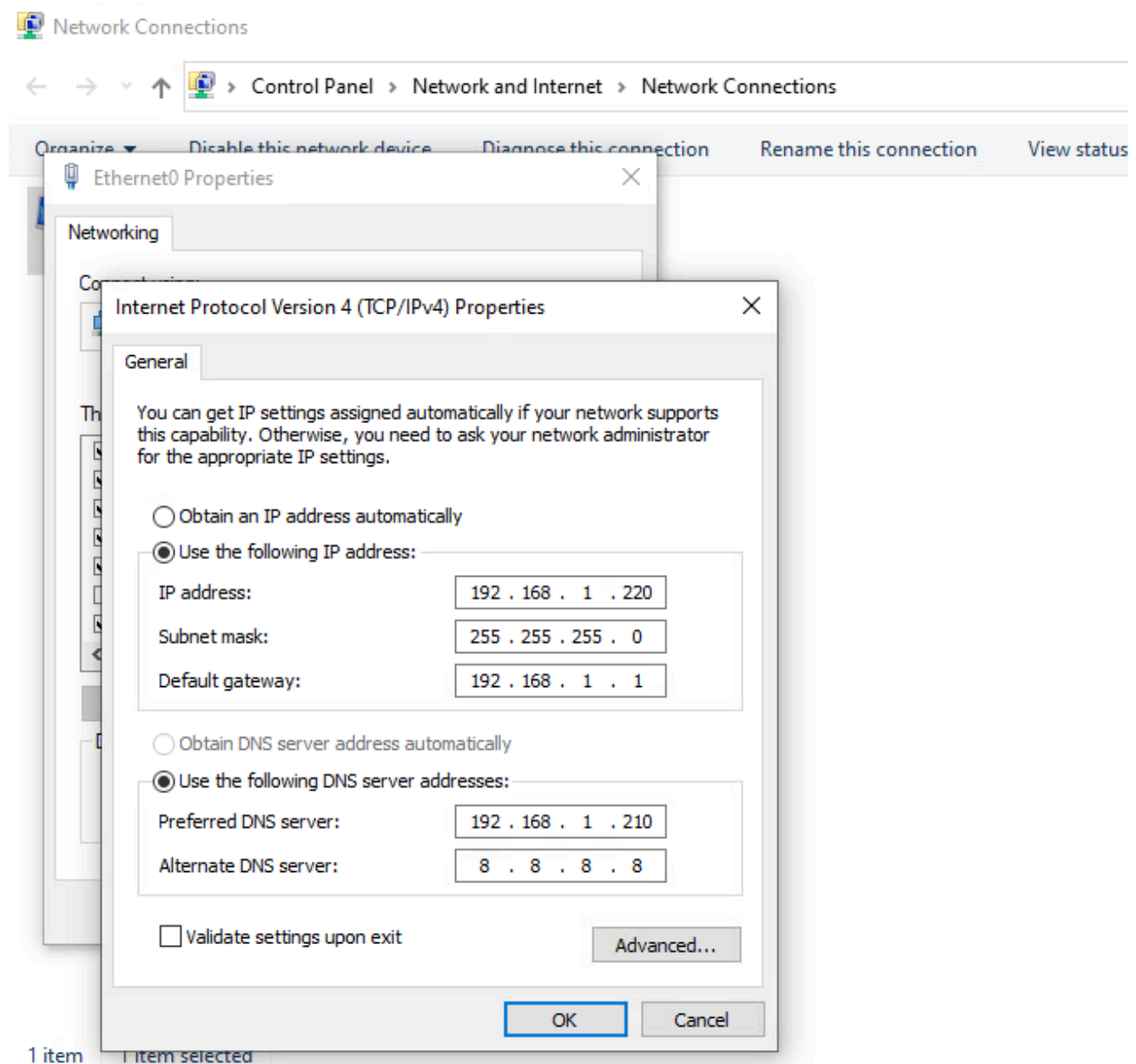


Figure 20: Networking settings for Windows 10

Click on the Windows icon, type in 'join domain' and open up 'Access work or school'. Click on the 'Connect' button and then click 'join this device to a local Active Directory domain'.

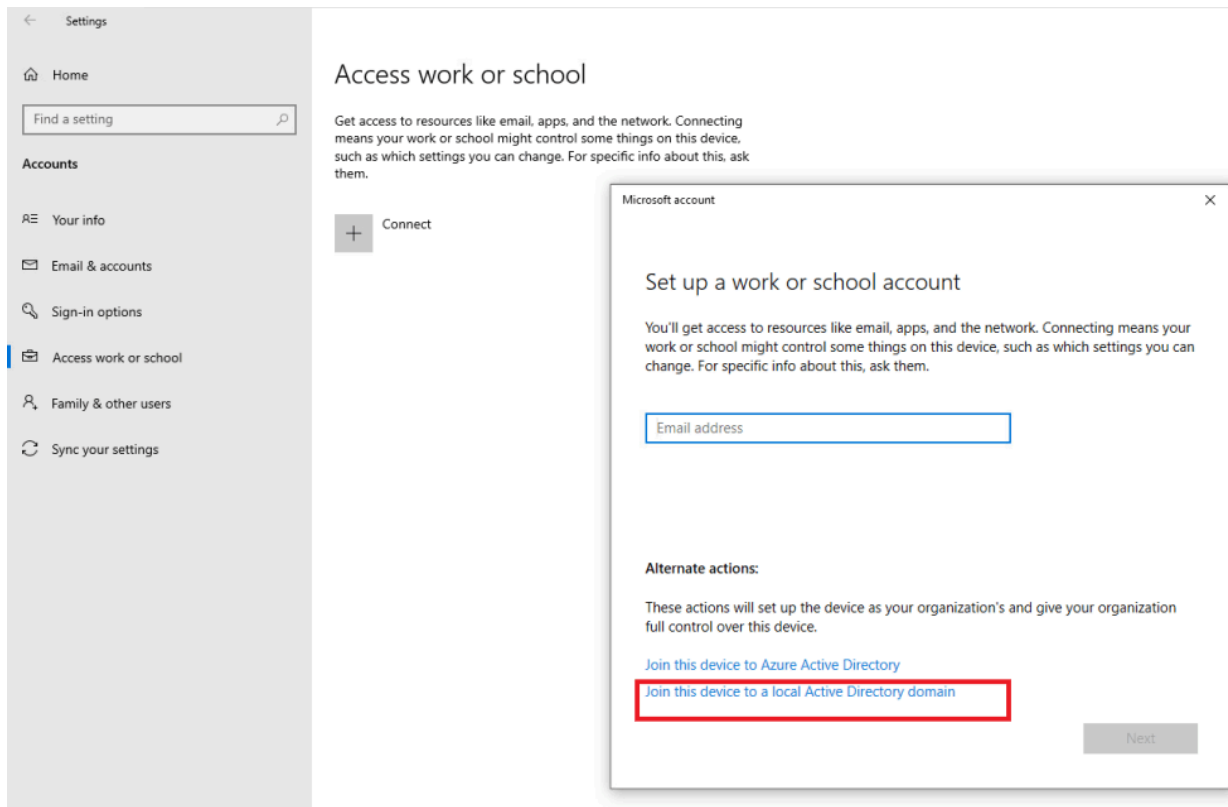


Figure 21: Select the highlighted box

Enter the FQDN (Fully qualified domain name) of your domain and click 'next'.



Figure 22: Enter the FQDN of your domain.

Note: If you get an error saying the domain was unable to be found, double check your DNS settings and ensure the Windows 10 machine can reach the Domain Controller.

You will then be prompted for credentials. This is where you will input your newly created Domain Administrator's credentials.

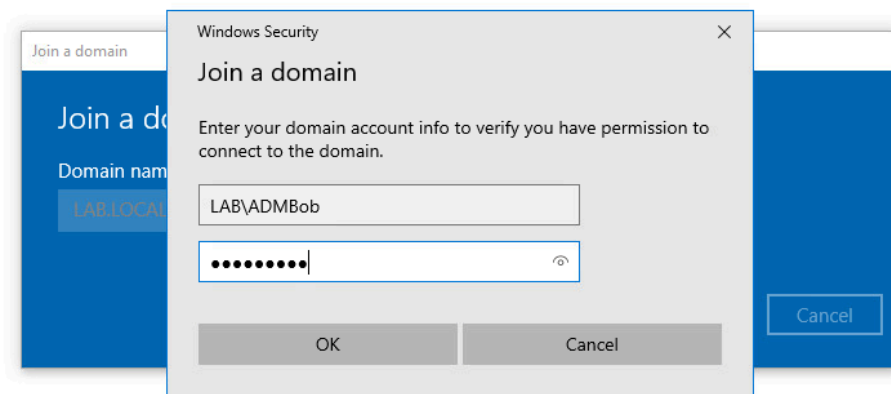


Figure 23: Enter your DA's credentials to join the domain

Reboot the PC and it will then be joined to the domain.

Winlogbeat

Now that we have a workstation and domain controller as well as an ELK server, we need to configure our two Windows machines to send logs to the ELK server. To do this, we need a program called '[Winlogbeat](#)'. In addition, I recommend also installing [Sysmon](#). Download the .zip file for Winlogbeat and unzip it to a folder inside one of the Windows machines. Open a PowerShell window and navigate to the Winlogbeat directory in PowerShell. Run the following command

```
Set-ExecutionPolicy bypass
```

Select [a] when prompted

Then run the script

```
.\install-service-winlogbeat.ps1
```

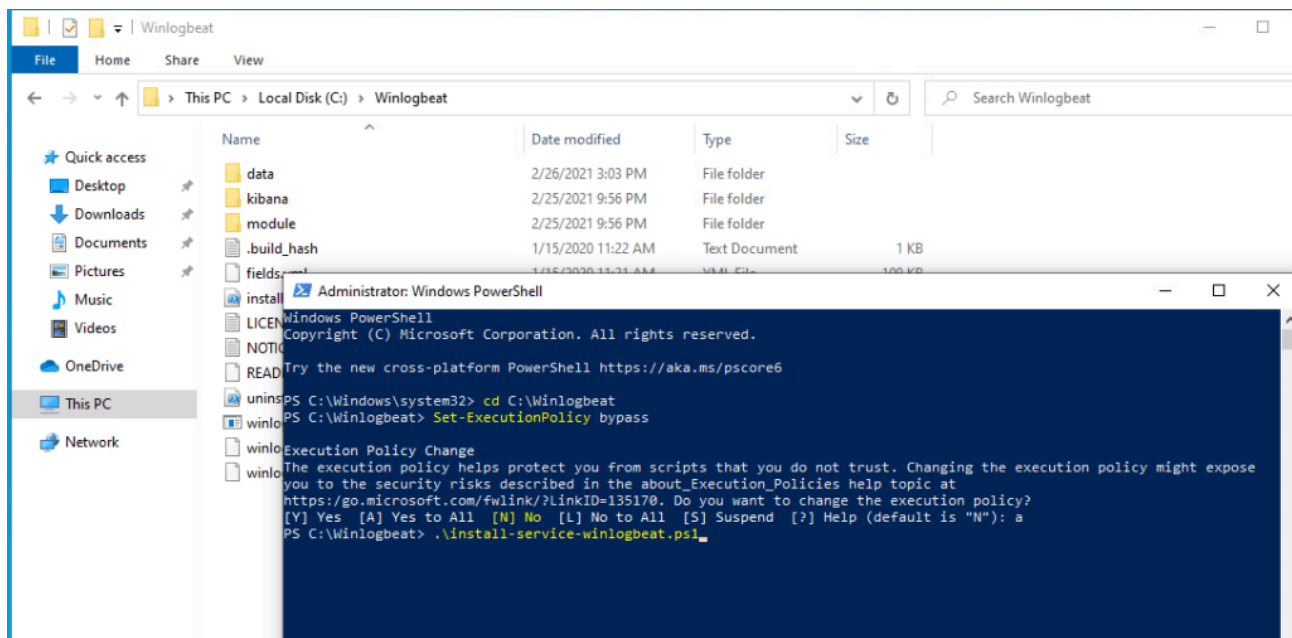


Figure 24: Installing Winlogbeat

Next, open “winlogbeat.yml” in Notepad. Copy+paste the following while changing the “hosts” IPs to match your ELK’s server IPs.

```
===== Winlogbeat specific options =====
winlogbeat.event_logs:
  name: Application
  ignore_older: 30m
  name: Security
  ignore_older: 30m
  name: System
  ignore_older: 30m
  name: Microsoft-windows-sysmon/operational
  ignore_older: 30m
  name: Microsoft-windows-PowerShell/Operational
  ignore_older: 30m
  event_id: 4103, 4104
  name: Windows PowerShell
  event_id: 400,600
  ignore_older: 30m
  name: Microsoft-Windows-WMI-Activity/Operational
  event_id: 5857,5858,5859,5860,5861
output.elasticsearch:
  hosts: ["ELKIPHERE:9200"]
  username: "elastic"
  password: "changeme"
setup.kibana:
  host: "ELKIPHERE"
```

Then start the winlogbeat service

Start-Service winlogbeat

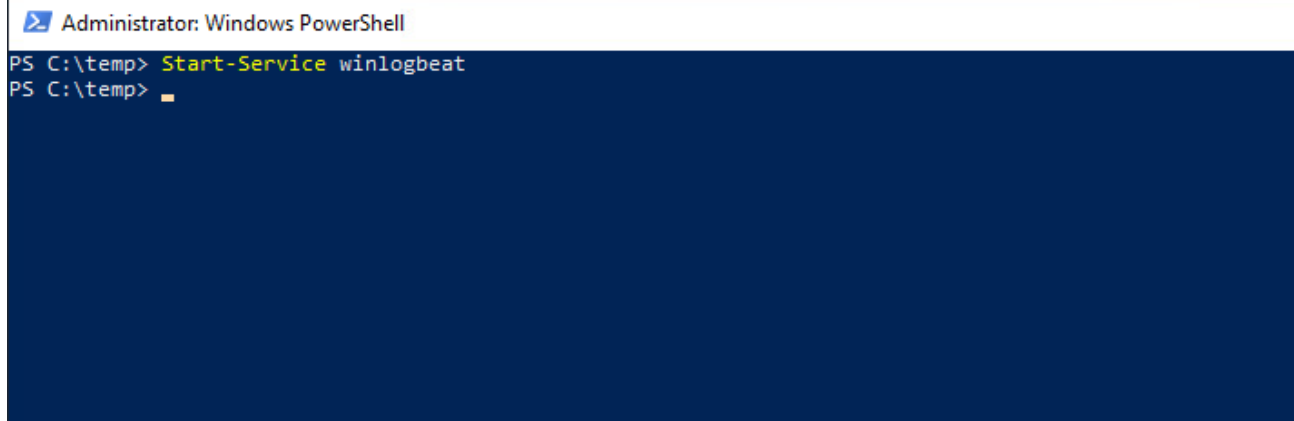


Figure 25: Starting winlogbeat

Once the service is started you can verify that the connection works by running

.\winlogbeat setup -e

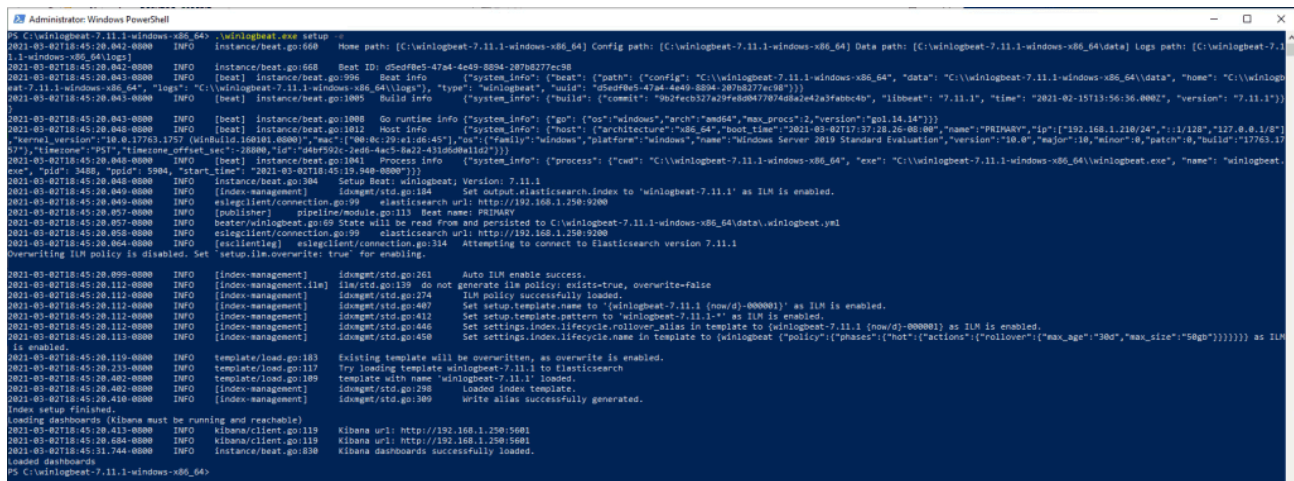


Figure 26: Checking winlogbeat config file

Click the drop down menu on the side and under 'Analytics' and go to 'Discover'. You should now be seeing Windows logs.

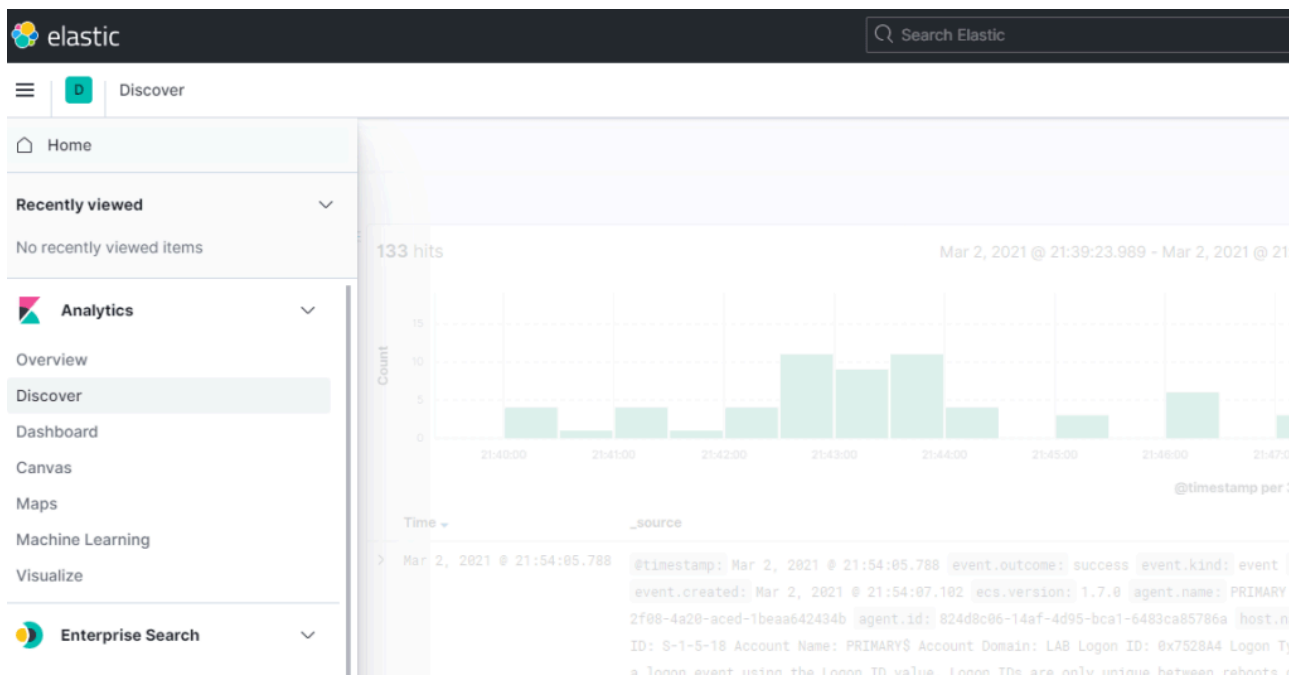


Figure 28: Viewing Windows Logs

Ensure that the time is synchronized properly within your lab as those are the times that will be reflected with the logs in Kibana. Otherwise, you can set your time filter to a different time.

Your basic detection lab is now ready to go! As said earlier, I recommend installing Sysmon on the Windows hosts to get detailed events out of them.